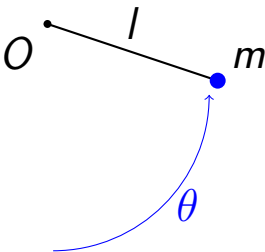


An Example of MOCK: The Simple Pendulum

Adam Handwerger



$$\ddot{\theta} + \left(\frac{g}{l}\right) \sin(\theta) = 0$$

State Space Representation

With $x(t) = [\theta(t), \dot{\theta}(t)]^\top$, $\omega_0 = \sqrt{\frac{g}{l}}$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ -\omega_0^2 \sin x_1(t) \end{bmatrix} \quad (\text{SS})$$

The Analytical Solution

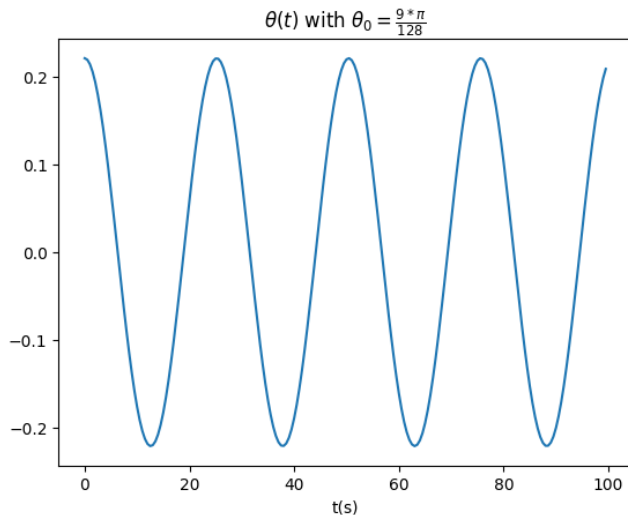
The special function sn used below is called the Jacobi elliptic function.

$$\theta(t) = 2 \sin^{-1} \left(\sin \frac{\theta_0}{2} \text{sn} \left[K \left(\sin^2 \frac{\theta_0}{2} \right) - \omega_0 t; \sin^2 \frac{\theta_0}{2} \right] \right)$$

with

$$K(m) = \int_0^1 \frac{dz}{\sqrt{(1-z^2)(1-mz^2)}}$$

For small θ_0 the pendulum is a harmonic oscillator



Synthesizing data for the MOCK training set

The state space system (SS) was numerical integrated to find $x(t)$ with `solve_ivp`, using the DOP853 method. From this, $\dot{\theta}(t)$ was used along with the analytical value $\theta(t)$

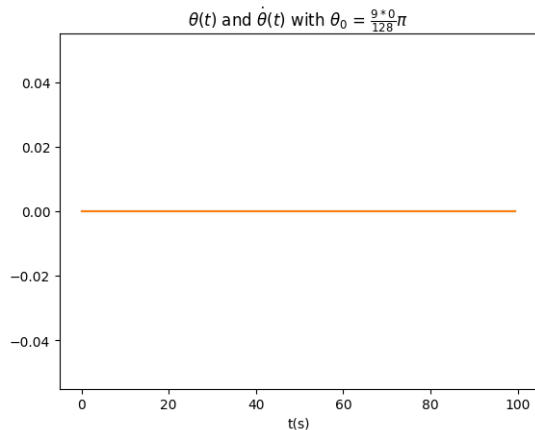
Initial Values

The various trajectories, $\theta_i(t)$ correspond to different θ_0 . For synthesizing the data a uniform distribution of initial positions from $\theta_0 = 0$ to $\theta_0 = \frac{127}{128}\pi$ were used. Only trajectories with zero initial velocity were considered.

Accuracy of Numerical Integration

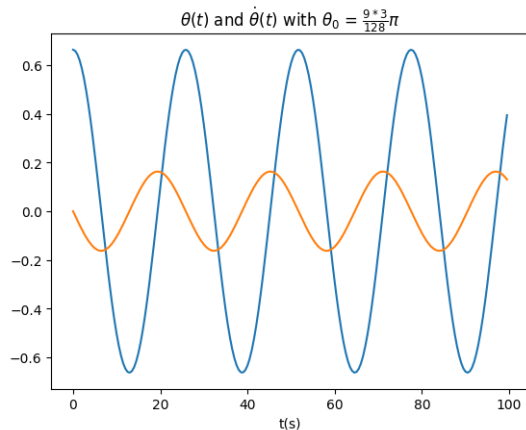
The percent error is defined as $\frac{\| \theta_n(t) - \theta_a(t) \|}{\| \theta_a(t) \|} \times 100$
where $\theta_a(t)$ is the analytic solution and $\theta_n(t)$ is the numerical solution.

Sample Trajectory from Validation Set



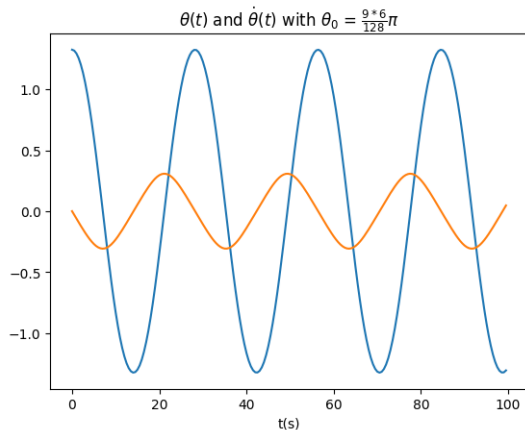
percent error=0.00

Sample Trajectory from Validation Set



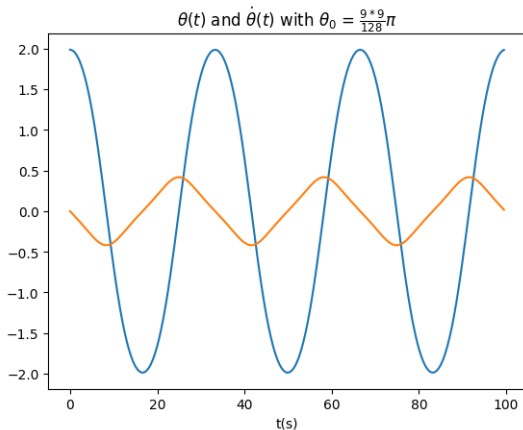
percent error=0.0018

Sample Trajectory from Validation Set



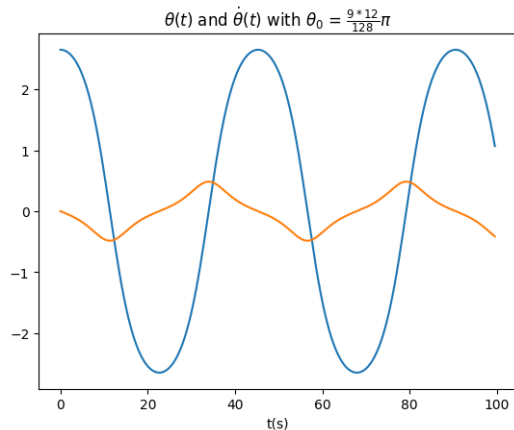
percent error=0.0010

Sample Trajectory from Validation Set



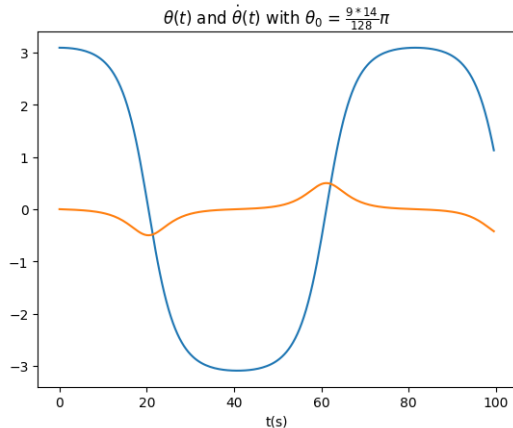
percent error=0.00092

Sample Trajectory from Validation Set



percent error=0.0019

Sample Trajectory from Validation Set



percent error=0.073

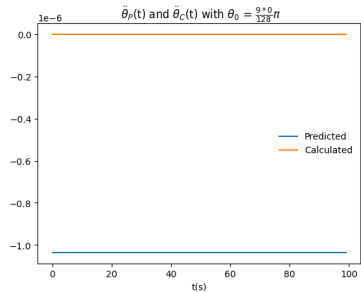
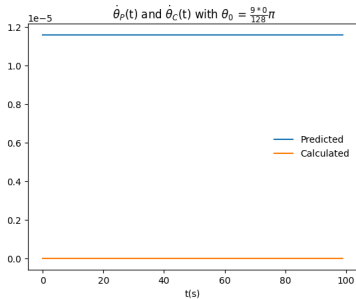
Tuning the Hyperparameters γ, λ with the Validation Set

For the validation set we choose

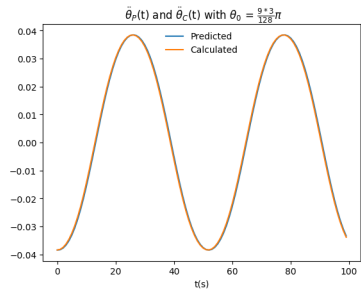
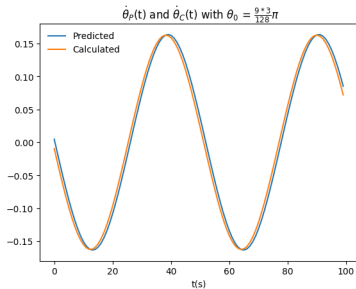
$$\theta_0 = \frac{9\pi * n}{128} \text{ for } n = 0, 1, \dots, 14$$

which gave $\gamma = 10$ and $\lambda = 5 * 10^{-7}$ as optimal.

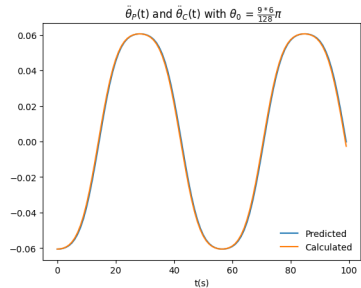
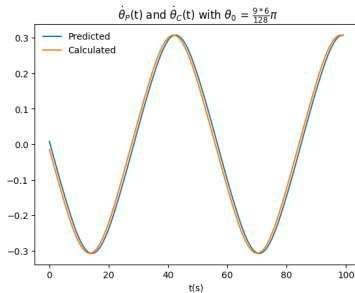
Slope Field of Sample Trajectories from Validation Set



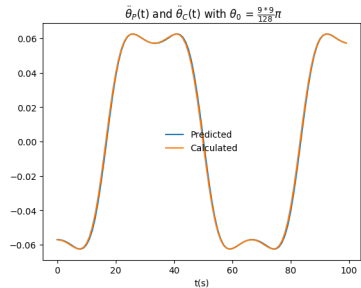
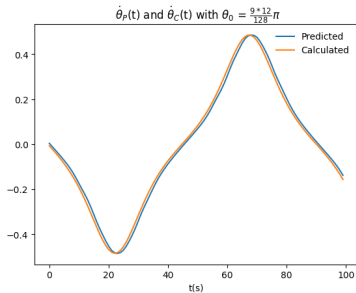
Slope Field of Sample Trajectories from Validation Set



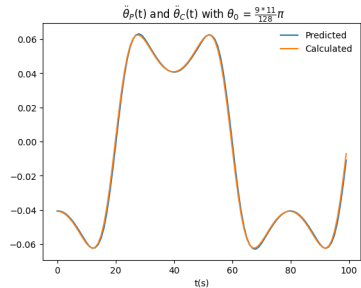
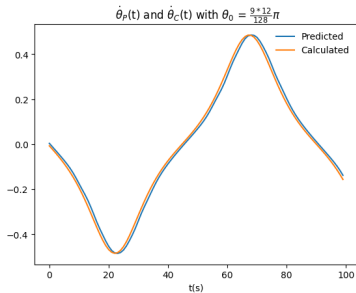
Slope Field of Sample Trajectories from Validation Set



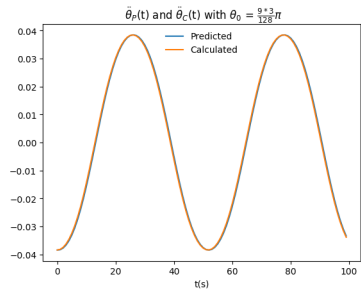
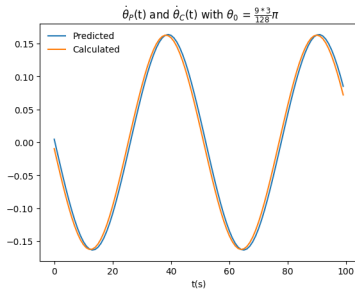
Slope Field of Sample Trajectories from Validation Set



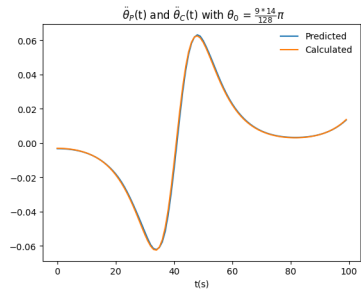
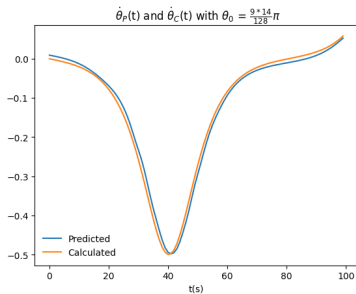
Slope Field of Sample Trajectories from Validation Set



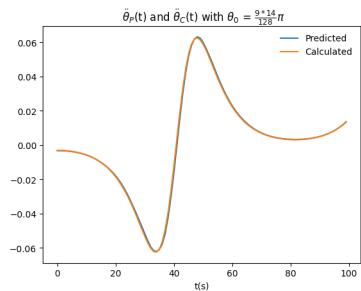
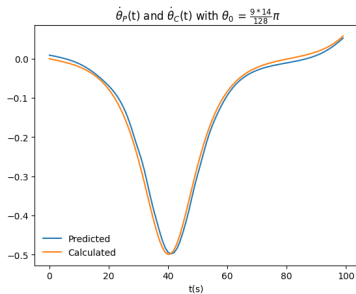
Slope Field of Sample Trajectories from Validation Set



Slope Field of Sample Trajectories from Validation Set



Slope Field of Sample Trajectories from Validation Set



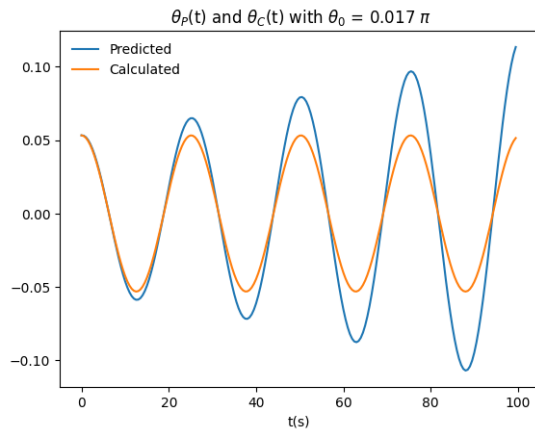
The Training Set

The training set consisted of 15 trajectories which were slightly different from the validation set. The values of θ_0 were evenly spaced and ranged from $.0007\pi$ to 0.985π .

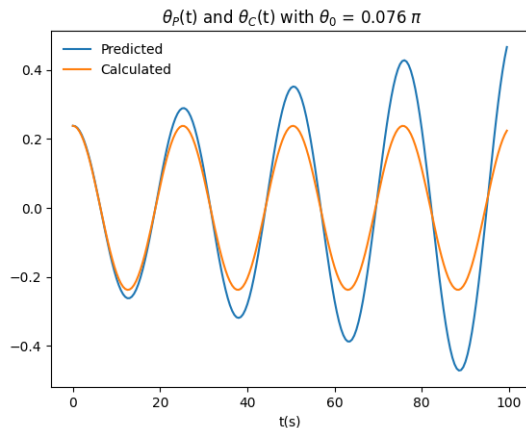
A Comparison of Trajectories as Predicted by MOCK vs. Analytical Derived

The slope field derived by the MOCK method was numerically integrated using the 'RK45' method for five random trajectories with $\theta_0 \in (0, \pi)$ and compared with more advanced methods available from the *solve_ivp* package. No noticable difference was seen in the results by switching integration methods.

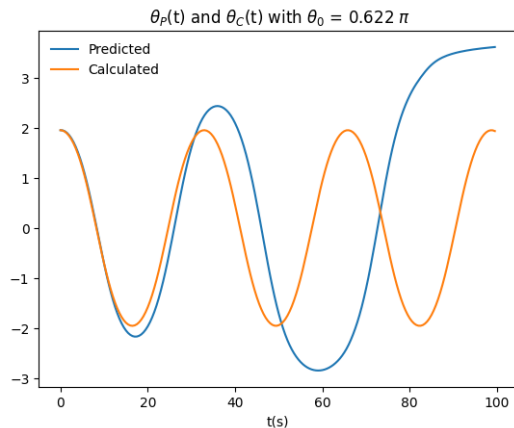
A Comparison of Trajectories



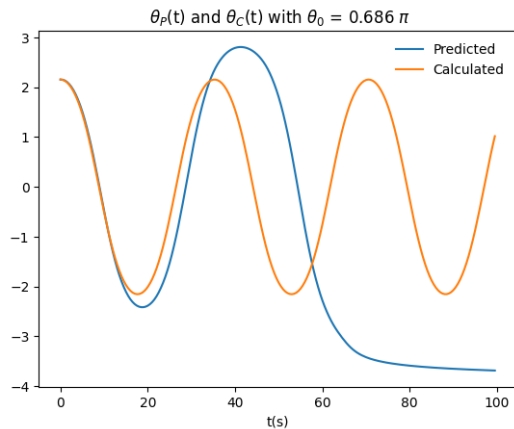
A Comparison of Trajectories



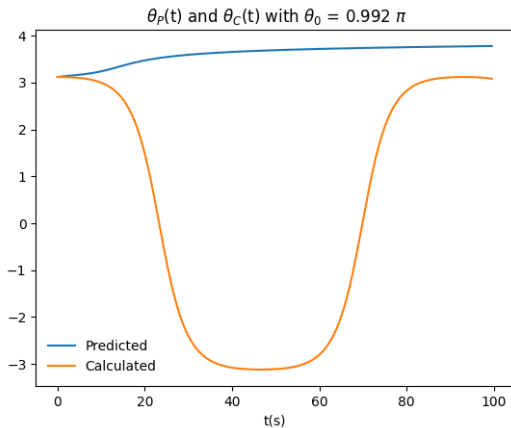
A Comparison of Trajectories



A Comparison of Trajectories



A Comparison of Trajectories



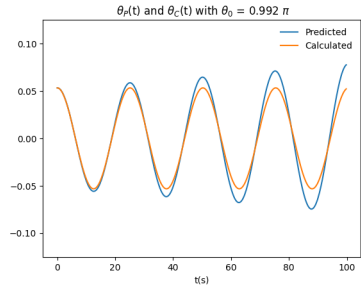
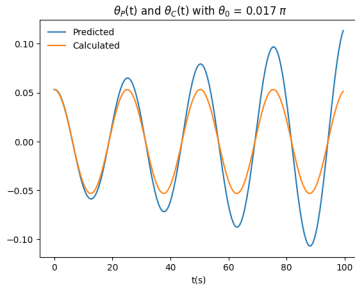
Possible Improvements to Modeling Trajectories

- ▶ Add more trajectories to the training set
- ▶ Increase the sampling rate for each trajectory in the training set

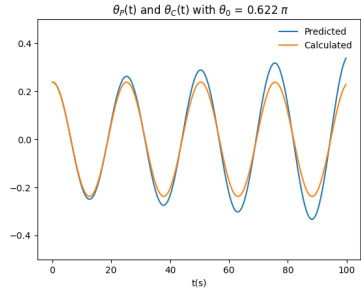
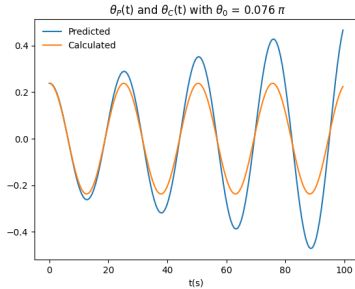
Results

Doubling the number of trajectories did not help with learning the training set nor with θ_0 chosen at random. Doubling of the sampling rate lead to a substainal improvement as shown below. This suggests that any pendulum trajectory can be predicted for all time, but only if the training set is sampled continuously.

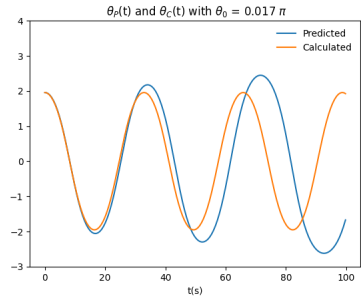
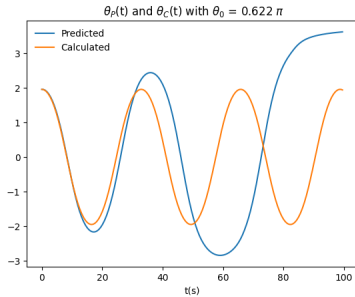
A Comparison of Different Sampling Rates



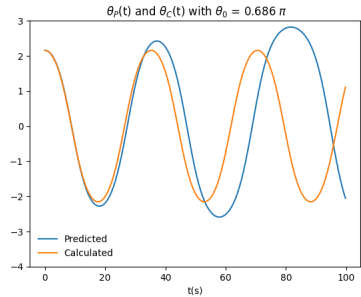
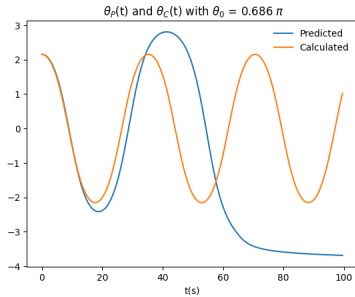
A Comparison of Different Sampling Rates



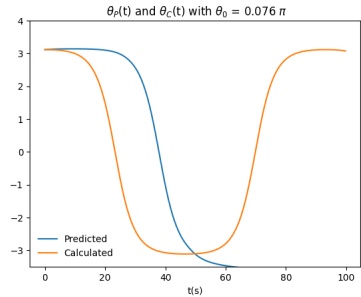
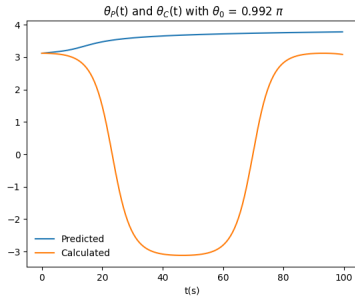
A Comparison of Different Sampling Rates



A Comparison of Different Sampling Rates



A Comparison of Different Sampling Rates



References

- (1) Rielly, V. et. all (2025, Jane 8) *MOCK: an Algorithm for Learning Nonparametric Differential Equations via Multivariate Occupation Kernel Functions*, arXiv:2306.10189v3

- (2) Beléndez, C. et. all (2007, Aug. 28) *Exact solution for the nonlinear pendulum*, Sociedade Brasileira de Física, v. 29 n.4 p.645-648